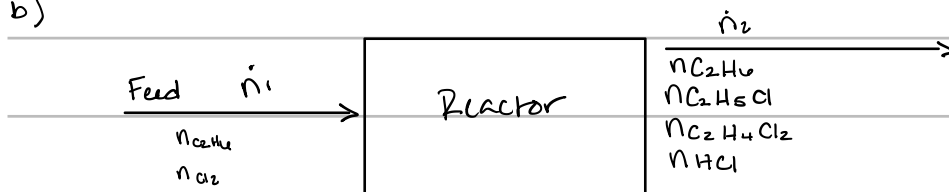


① a) To maximize for selectivity, you would design for low conversion.

The undesired second product consumes C_2H_5Cl which is produced in the first chlorination.

If we aimed for high conversion, more C_2H_5Cl would be produced meaning more undesired product. Low conversion means less product, but also less undesired product.

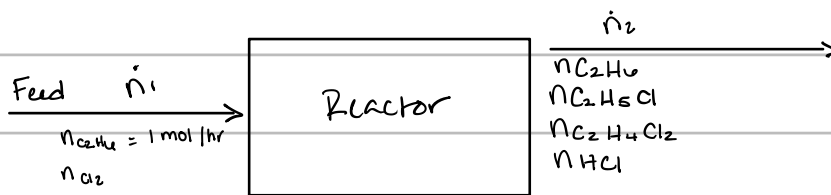
b)



Could also say that we want to have C_2H_6 in excess

DoF: $4a) 0 + 4b) 1 - 5a) 0 + 5b) 0 + 5c) 0 + 5d) 5 \rightarrow 7 - 5 = 2$ underspecified

c)



15% ethane reacted

14 moles C_2H_5Cl / 1 mole $C_2H_4Cl_2$

$$nC_2H_5Cl + nC_2H_4Cl_2 = 0.15 nC_2H_6 \quad \& \quad nC_2H_5Cl = 14 nC_2H_4Cl_2$$

$$14 nC_2H_4Cl_2 + nC_2H_4Cl_2 - 0.15 nC_2H_6 \rightarrow nC_2H_4Cl_2 = 0.01 nC_2H_6 \quad nC_2H_5Cl = 0.14 nC_2H_6$$

$$\frac{Cl_2}{C_2H_6} = \frac{0.15 nC_2H_6}{nC_2H_6} = 0.15 \quad \text{yield} = \frac{nC_2H_5Cl}{nC_2H_6} = 0.14 \times 100 = 14\%$$

-10 pts

d) 14% conversion, no Cl_2 , new heavier dichloro species

The presence of a new, heavier dichloro could be explained by over-chlorination. Chlorination can go beyond the dichloro stage, producing tri or tetra chlorinated products. These will be heavier than dichloro products.

2) $C_3H_8O_3 + 3H_2O \rightleftharpoons 3CO_2 + 7H_2$

$$P = 1.2 \text{ atm} \quad T = 200^\circ C \quad K_a = 55 \text{ atm}^4$$

5 mol H_2O : 1 mol gly.

$\dot{n}_g = 150 \text{ g/h}$



$$\dot{n}_g = \frac{150 \text{ g/h}}{92.09 \text{ g/mol}} = 1.629 \frac{\text{mol}}{\text{h}}$$

5:1 ratio so $\dot{n}_w = 5(1.629) = 8.145 \frac{\text{mol}}{\text{h}}$

$$C_3H_8O_3: \dot{n}_{g,out} = \dot{n}_{g,in} - \xi \quad \text{total: } \dot{n}_{total,out} = \dot{n}_g + \dot{n}_w + 6\xi = 9.774 + 6\xi$$

$$H_2O: \dot{n}_{w,out} = \dot{n}_{w,in} - 3\xi \quad X_{H_2} = 0.54 \text{ of } \dot{n}_{total,out} \text{ so } X_{H_2} = \frac{7\xi}{\dot{n}_{out}} \rightarrow 7\xi = 0.54(9.774 + 6\xi) \rightarrow \xi = 1.404 \frac{\text{mol}}{\text{h}}$$

$$CO_2: \dot{n}_{CO_2,out} = \dot{n}_{CO_2,in} + 3\xi \quad f_c = \frac{\xi}{\dot{n}_g} = \frac{1.404}{1.629} \times 100 = 86.2\% \text{ frac. conversion of glycerol}$$

$$H_2: \dot{n}_{H_2,out} = \dot{n}_{H_2,in} + 7\xi$$

$$\text{mol fractions} = \frac{(\dot{n}_i - \xi)}{\dot{n}_{total}}$$

$$X_g = \frac{0.225}{18.198} = 0.0124$$

$$X_w = \frac{3.433}{18.198} = 0.2111$$

$$X_{CO_2} = \frac{4.212}{18.198} = 0.2314$$

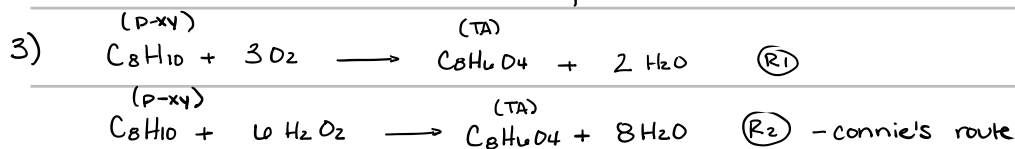
$$X_{H_2} = \frac{9.828}{18.198} = 0.54$$

$$K_p = \frac{(0.277)^3 (0.648)^7}{(0.01483)(0.2593)^3} = 3.98 \text{ atm}^0 \quad \text{since this is } \ll \text{ than } K_a = 55, \text{ not equilibrium}$$

$$\text{At equilibrium?} \quad 55 = \frac{(3\dot{\xi}/n)^3 (7\dot{\xi}/n)(1.2)}{((\dot{n}_g - \dot{\xi})/n)((\dot{n}_w - 3\dot{\xi})/n)^3} \quad \text{so } \dot{\xi} = 1.574 \frac{\text{mol}}{n} \quad \text{at eq pt.}$$

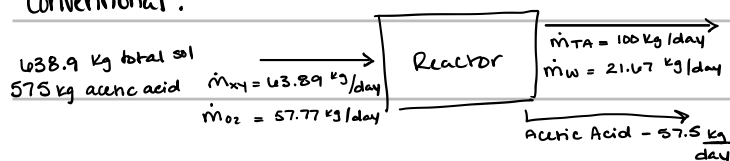
$$X_{eq} = \frac{1.574}{1.629} \times 100 = 96.7\% \quad \checkmark$$

$$\dot{n}_{H_2} = 7\dot{\xi} = 7(1.574) = 11.018 \frac{\text{mol}}{n}, \quad \dot{n}_{total} = 9.774 + 6\dot{\xi} = 19.218 \frac{\text{mol}}{n}, \quad X_{H_2} = \frac{11.018}{19.218} \times 100 = 57.4\% \quad \checkmark$$



Basis: 100 kg/day TA $\rightarrow$ 100000 g/day

Conventional:



$$\dot{n}_{TA} = \frac{100000g}{166.13g/mol} = 601.8 \text{ mol/day} \quad | \text{ TA : 1 p-xy } \text{ so } 601.8 \frac{\text{mol}}{\text{day}} = \dot{n}_{xy} \text{ too}$$

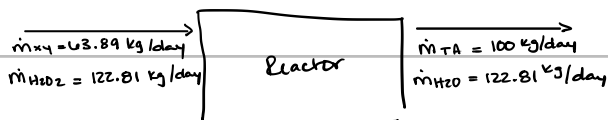
$$\text{mass p-xy required: } \dot{m}_{xy} = 601.8 \cdot 106.16g/mol = 63.89 \text{ kg/day}$$

$$\dot{n}_{O_2} = 3(601.8) = 1805.4 \frac{\text{mol}}{\text{day}} \rightarrow \dot{m}_{O_2} = (1805.4)(32g/mol) = 57.77 \text{ kg/day} \quad \left. \vphantom{\dot{n}_{O_2}} \right\} \rightarrow \dot{m}_{total, in} = 63.89 + 57.77 = 121.66 \text{ kg/day}$$

$$\dot{n}_{H_2O} = 2(601.8) = 1203.6 \frac{\text{mol}}{\text{day}} \rightarrow \dot{m}_{H_2O} = (1203.6)(18.02g/mol) = 21.67 \text{ kg}$$

$$\text{Atom economy: } AE = \frac{100 \text{ kg}}{121.66 \text{ kg}} \times 100 = 82.2\% \quad \checkmark$$

Connie's:



$$\dot{m}_{xy} = 63.89 \text{ kg/day}, \quad \dot{n}_{H_2O_2} = 6(601.8) = 3610.8 \frac{\text{mol}}{\text{day}} \rightarrow \dot{m}_{H_2O_2} = (3610.8)(34.014g/mol) = 122.81 \text{ kg/day}$$

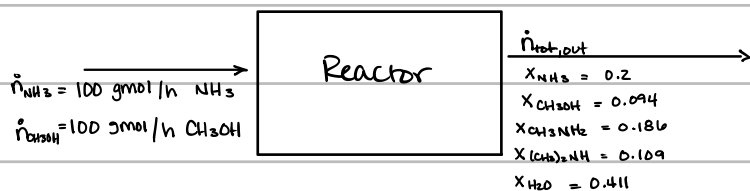
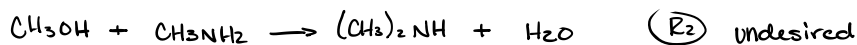
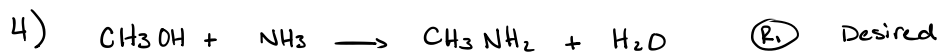
$$\dot{n}_{H_2O} = 8(601.8) = 4814.4 \frac{\text{mol}}{\text{day}} \rightarrow \dot{m}_{H_2O} = (4814.4)(18.02g/mol) = 186.7 \text{ kg/day} \quad \dot{m}_{total, in} = 186.7 \text{ kg/day}$$

$$AE_{connie} = \frac{100 \text{ kg}}{186.7 \text{ kg}} \times 100 = 53.6\% \quad \checkmark \text{ - Connie's atom economy is much lower than conv. process.}$$

Connie's idea does show good selectivity & avoids acetic acid solvent.

This could be good in some cases, but maybe would not prefer it over the conventional method in a generic case like this. $\checkmark$

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$$\text{CH}_3\text{OH}: \dot{n}_{\text{CH}_3\text{OH}} - \dot{\xi}_1 - \dot{\xi}_2 = 0.094 \dot{n}_{\text{total, out}} = 100 - \dot{\xi}_1 - \dot{\xi}_2 = 0.094 \dot{n}_{\text{total, out}}$$

$$\text{NH}_3: \dot{n}_{\text{NH}_3} - \dot{\xi}_1 = 0.20 \dot{n}_{\text{total, out}} = 100 - \dot{\xi}_1 = 0.20 \dot{n}_{\text{total, out}}$$

$$\text{CH}_3\text{NH}_2: \dot{\xi}_1 - \dot{\xi}_2 = 0.186 \dot{n}_{\text{total, out}}$$

$$(\text{CH}_3)_2\text{NH}: \dot{\xi}_2 = 0.109 \dot{n}_{\text{total, out}}$$

$$\text{H}_2\text{O}: \dot{\xi}_1 + \dot{\xi}_2 = 0.411 \dot{n}_{\text{total, out}}$$

$$\dot{\xi}_2 = 0.109 \dot{n}_{\text{tot}} \rightarrow \dot{\xi}_1 = 0.186 \dot{n}_{\text{tot}} + \dot{\xi}_2 = 0.186 \dot{n}_{\text{tot}} + 0.109 \dot{n}_{\text{tot}} = 0.295 \dot{n}_{\text{tot}}$$

from
NH₃ balance: $100 = 0.495 \dot{n}_{\text{tot}} \rightarrow \dot{n}_{\text{tot}} = 202.0 \text{ g mol/h}$

so, $\dot{\xi}_1 = 59.6 \frac{\text{g mol}}{\text{h}}$ ✓ $\dot{\xi}_2 = 22.0 \frac{\text{g mol}}{\text{h}}$

$$X_{\text{MeOH}} = \frac{\dot{\xi}_1 + \dot{\xi}_2}{100} = 0.816 \times 100 = 81.6\% \text{ methanol converted} \quad \checkmark$$

$$\text{Yield}_{\text{MeOH}} = \frac{\dot{\xi}_1 + \dot{\xi}_2}{100} = 0.376 \times 100 = 37.6\% \text{ yield} \quad \times -2.5$$

$$\text{Selectivity}_{\text{MeOH}} = \frac{\text{mols CH}_3\text{NH}_2 \text{ formed}}{\text{mols (CH}_3)_2\text{NH formed}} = \frac{\dot{\xi}_1 + \dot{\xi}_2}{\dot{\xi}_2} = \frac{59.6 + 22}{22} = 3.62 \text{ mol CH}_3\text{NH}_2 \text{ per mol (CH}_3)_2\text{NH} \quad \times -2.5$$

Reaction Rates: $R_1 = \dot{\xi}_1 = 59.6 \frac{\text{mol}}{\text{h}} \quad \& \quad R_2 = \dot{\xi}_2 = 22.0 \frac{\text{mol}}{\text{h}}$

At equilibrium? $R_1: K_1 = 4.0 \quad \text{so, } K_{1, \text{ex}} = \frac{(X_{\text{CH}_3\text{NH}_2})(X_{\text{H}_2\text{O}})}{(X_{\text{CH}_3\text{OH}})(X_{\text{NH}_3})} = \frac{(0.186)(0.411)}{(0.094)(0.20)} = 4.07 \quad \text{close! } \checkmark$

$R_2: K_2 = 2.5 \quad \text{so, } K_{2, \text{ex}} = \frac{(X_{(\text{CH}_3)_2\text{NH}})(X_{\text{H}_2\text{O}})}{(X_{\text{CH}_3\text{OH}})(X_{\text{CH}_3\text{NH}_2})} = \frac{(0.109)(0.411)}{(0.094)(0.186)} = 2.56 \quad \text{close! } \checkmark$

At eq pt: increasing NH₃:CH₃OH ratio favors monomethylamine

increasing temp. favors endothermic reaction (R₂) so low temps improve CH₃NH₂ selectivity

max methylamine conc: high NH₃, low T

max methylamine : dimethylamine ratio: excess NH₃ ✓

max yield of methylamine based on ammonia: NH₃ close to stoichiometric value

max yield based on methanol: use a little excess NH₃

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